METRIC
MIL-PRF-8565/6B(AS)
19 July 2012
SUPERSEDING
MIL-PRF-8565/6A(AS)
25 June 2001

PERFORMANCE SPECIFICATION SHEET

# BATTERY, STORAGE, AIRCRAFT, MAINTENANCE-FREE, 24-VOLT, 1.5-AMPERE-HOUR

This specification is approved for use by the Naval Air Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-8565.

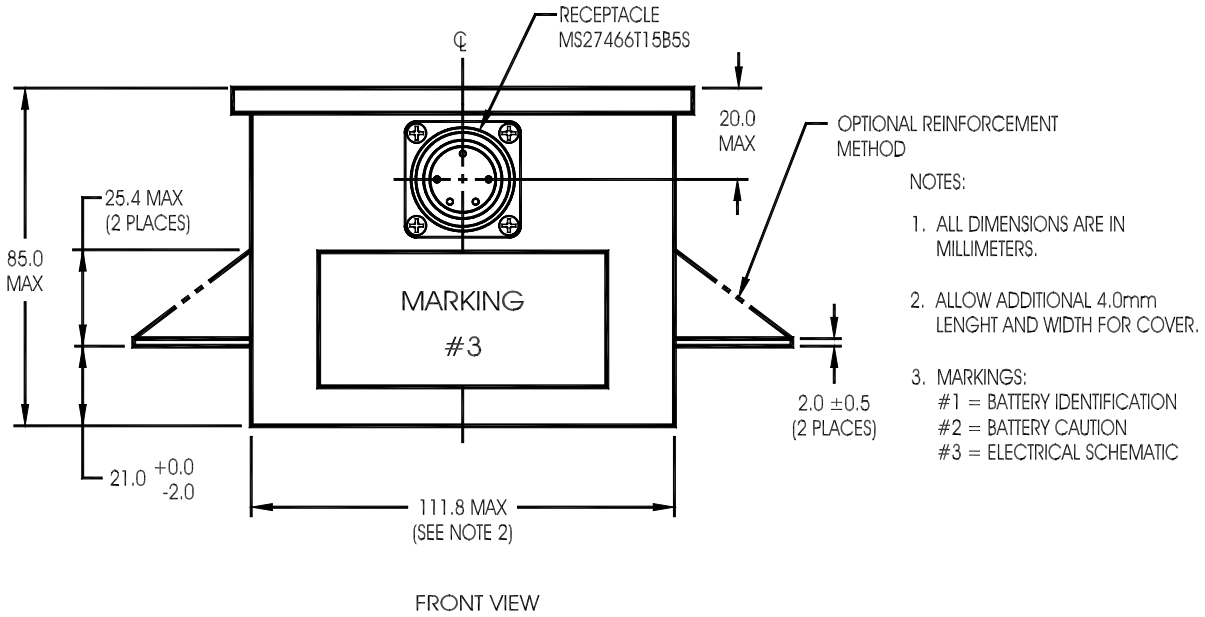



FIGURE 1. Dimensions and configuration for PIN D8565/6-1.



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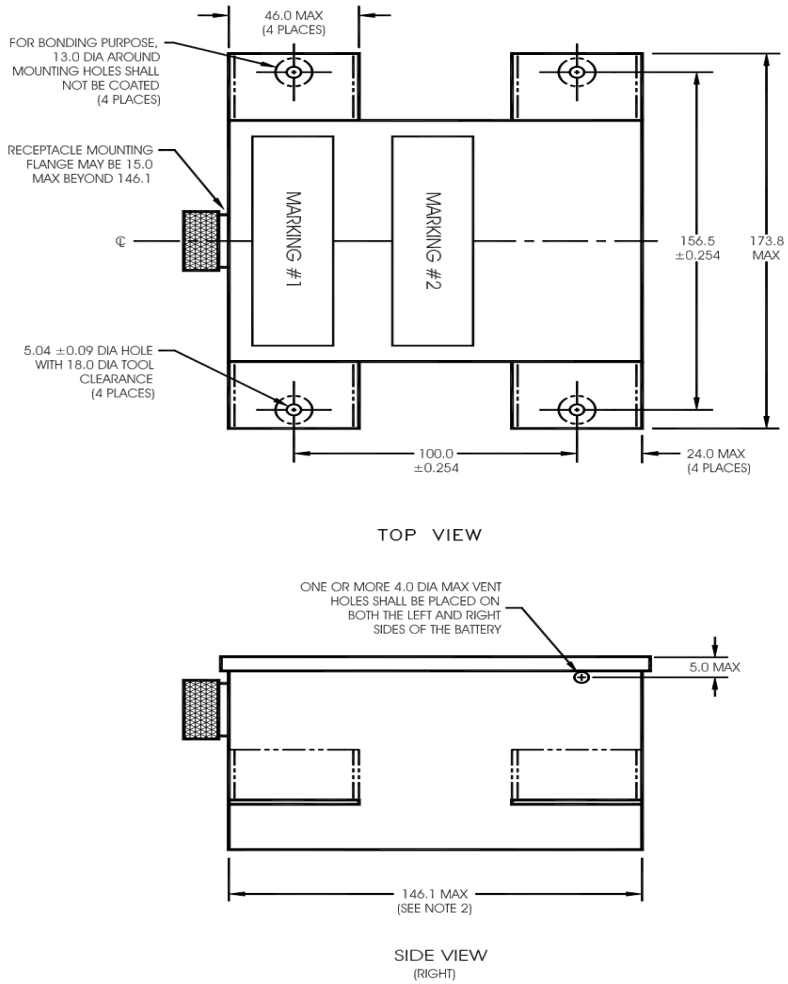



FIGURE 1. Dimensions and configuration for PIN D8565/6-1 – Continued.



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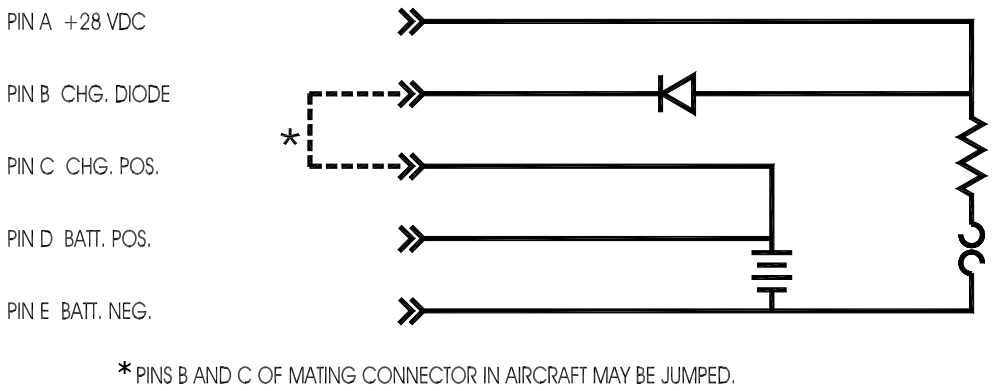



FIGURE 2. Electrical schematic for marking #3.

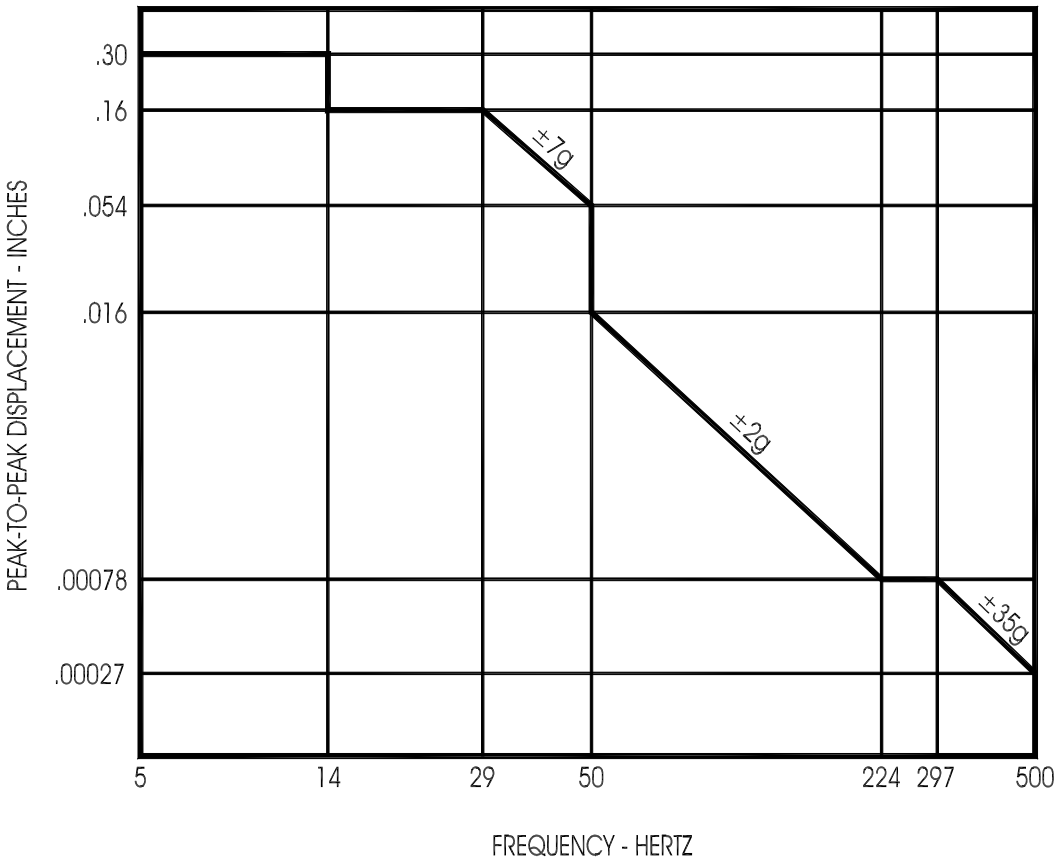



FIGURE 3. Sine wave vibration test levels.



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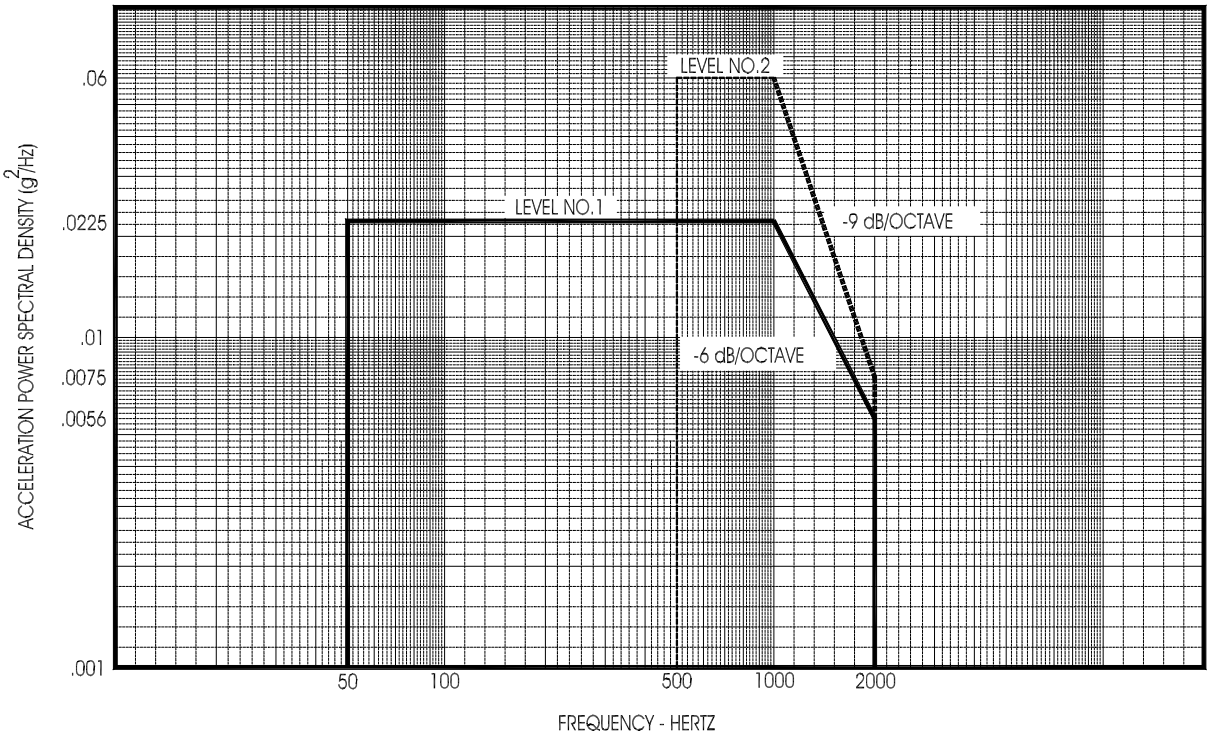



FIGURE 4. Random wave vibration test levels.

REQUIREMENTS:

1. Part or identifying number. The part or identifying number (PIN) of the battery covered by this specification sheet shall be D8565/6-1.

2. Dimensions and weight. The dimensions of the battery shall be as shown on figure 1 contained in the specification sheet. The weight of the battery shall not be greater than 2.9 kilograms (see 3.8.3 and 4.5.3 of MIL-PRF-8565).

3. Capacity. The rated capacity of the battery shall be not less than 1.5 ampere-hours (Ah). See 3.8.6 as modified herein.

4. Aircraft installation. For installation in the aircraft, the battery shall be supported as shown on figure 3 of MIL-PRF-8565. Vibration isolation may be provided to ensure that the vibration level does not exceed the levels of the 3-year accelerated life test of 4.5.10. If vibration isolators are used, sufficient sway clearance between the battery and the mounting structure shall be provided.



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5. Service life. The battery's service life (useful life) in aircraft shall be not less than 3 years. The battery shall require no scheduled maintenance during the 3 years (see 3.8.9, 4.5.10, and 6.5.10 of MIL-PRF-8565).

6. Qualification inspection and conformance inspection. Table I lists the required inspections.

7. MIL-PRF-8565 variance. The battery shall comply with MIL-PRF-8565, except as follows:

7.1 The following paragraphs and figures are not applicable:

3.6.2 Polarity marking.
3.8.5 Vent tubes.
3.8.7 Constant voltage discharge.
3.8.8 Strength of receptacle.
4.5.5 Strength vent tubes test.
4.5.8 Constant voltage discharge.
4.5.9 Strength of receptacle test.
4.5.12 Charge and discharge test at low temperature.
4.5.30 Wormgear retaining pin hardness.

7.2 Insert after the last sentence of the following paragraphs: "The battery shall also meet the requirements of 3.8.18."

3.8.11 Temperature rise and float.
3.8.12 Battery gas emission.
3.8.13.1 Ground storage.
3.8.14 Impact resistance (nonmetallic battery container only).

7.3 Insert into the following paragraphs: "Perform the test of 4.5.32. Examine the battery for the requirements of 3.8.18."

4.5.13 Discharge while inverted. Before the first sentence of step 4.5.13.a.
4.5.14 Altitude test. Before the first sentence of step 4.5.14.a.
4.5.15 Shock test (basic design). Before the first sentence of step 4.5.15.a.
4.5.16 Temperature shock test. Before the first sentence of 4.5.16.a.
4.5.18 Battery gas emission test. Before the first sentence of step 4.5.18.a.
4.5.20 Humidity test. Before the first sentence of step 4.5.20.a.
4.5.21 Salt fog test. Before the first sentence of step 4.5.21.a.
4.5.24 Impact resistance test (nonmetallic battery container). Before the first sentence of step 4.5.24.a.



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4.5.31 Final examination. Before the first sentence.

7.4 Modify the following paragraphs:

3.5.4 Venting. Delete the entire text and insert: "Each battery shall include two vent holes, located as shown on figure 1 of the specification sheet. Air shall flow freely into and through any vent hole, across the cells and exit through the other vent hole."

3.5.5 Receptacles. Delete the second sentence and insert: "The battery's electrical interface to the aircraft shall be the MS27466T15B5S receptacle. It shall be positioned and electrically connected as shown on figure 1 of the specification sheet.

a. Pin A shall be +28 VDC input for battery heater and optional charging.

b. *Pin B shall be charging diode output.

c. *Pin C shall be battery charge input.

d. Pin D shall be battery positive output.

e. Pin E shall be battery negative.

*Pins B and C of mating connector in aircraft installation may be jumpered if the optional charging from pin A is used."

3.5.10 Heaters. In 3.5.10.a, delete the text and insert the following: "Heaters shall be powered by 28.25 VDC in accordance with MIL-STD-704, accessible at connector pin A." In 3.5.10.b, delete the text and insert the following: "Heater current shall be not greater than 2.0 amperes at 28.25 VDC. The rated current of each heater control device shall be compatible with the heater current." In 3.5.10.h, delete "75° C (167° F)" and insert "60° C (140° F)."

3.6.4 Battery caution marking. In the second sentence, delete "6.3 mm" and insert "3.8 mm."

3.6.5 Marking for batteries with heaters. In the first sentence, delete "the battery identification marking and battery caution marking, " and insert, "the battery cover and the surface bearing the battery identification." At the end of the third sentence, insert: "(see figure 2 of the specification sheet).



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3.7 General requirements. In the second sentence of step e, delete “tubes” and insert “holes.” In the third sentence of step e, delete “ports” and insert “holes.”

3.8.6 Capacity and electrical performance. Delete the text and insert: “The rated capacity of the battery shall be not less than 1.5 ampere-hours. To meet the capacity requirement, each battery shall produce a potential of not less than 20.0 volts at not less than 60 minutes when connected to a 1.5-ampere load after being tested in accordance with 4.5.7 at 21° C (70° F) (1.5Ah/1.0hr/21° C/20.0 V at end of life). The battery 1C-rate shall be 1.5 amperes (see requirement 3 of the specification sheet). The battery shall also meet the requirements of 3.8.18.”

3.8.9 Cycling. Insert the following after the first sentence: “The battery potential shall be not less than 20.0 volts after the discharge at the end of each cycle. The battery shall also meet the requirements of 3.8.18.”

3.8.16 Output performance. Delete the text and insert: “The battery shall produce a potential of not less than 20.0 volts during and after the test of 4.5.28. The battery shall also meet the requirements of 3.8.18.”

3.8.17 Cold and hot temperature charge. Delete the text.

7.5 Add the following paragraphs:

“3.8.17.1 Half-hour charging. The battery shall produce not less than 1.25 ampere-hours after being tested in accordance with 4.5.29.1.c and 4.5.29.1.h. The battery shall also meet the requirements of 3.7 and 3.8.18.”

“3.8.17.2 One-hour charging. The battery shall produce not less than 1.25 ampere-hours after being tested in accordance with 4.5.29.2.e. The battery shall not sustain any damage when a 28.25 ± 0.5 volt charging source is applied continuously to the battery during the test of 4.5.29.2. The battery shall also meet the requirements of 3.7 and 3.8.18.”

“3.8.18 Diode resistance. The battery charging diode shall not be shorted or open after the test of 4.5.32. The ratio of the charging diode’s forward direction resistance to its reverse direction resistance shall be not less than 1:100.”

“3.8.19 Battery high rate discharge. The battery potential shall be not less than 18.0 volts during the test of 4.5.33. The battery shall also meet the requirements of 3.7 and 3.8.18.”

7.6 Modify the following paragraphs:



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4.5.6 Electrical test preparation. In the first sentence, change “18.0 volts” to “20.0 volts.”

4.5.7 Capacity discharge. Delete the first two sentences and insert: “Perform the test of 4.5.32. Examine the battery for the requirements of 3.8.18. Stabilize the battery at room temperature (see 4.4.1). Discharge the battery at the 1.0 C-rate to a cutoff voltage (see 6.15.4) of 20.0 volts.”

4.5.10 Cycling test. Delete the text and insert: “Perform the test of 4.5.32. Examine the battery for the requirements of 3.8.18. Constant potential charge the battery in accordance with 4.4.2 at room temperature (see 4.4.1). Subject the battery to 85 cycles of discharge and charge under the ambient conditions of 4.4.1. Each cycle shall consist of a 2.0-minute, 15-ampere discharge, followed immediately by a 118-minute constant potential charge in accordance with 4.4.2, followed by a 1.0-hour rest period. Examine the battery for the requirements of 3.8.9.”

4.5.11 Capacity discharge at 49° C (120° F). Insert before the first sentence: “Perform the test of 4.5.32. Examine the battery for the requirements of 3.8.18.” In the next to the last sentence, change “18.0 volts” to “20.0 volts.”

4.5.17 Temperature rise and float test. In step 4.5.17.a, insert before the first sentence: “Perform the test of 4.5.32. Examine the battery for the requirements of 3.8.18.” In the second sentence of step 4.5.17.e, change “18.0 volts” to “20.0 volts.”

4.5.19 Vibration test. Delete steps 4.5.19.a, 4.5.19.d, and 4.5.19.e. Insert the following:

“a. Perform the test of 4.5.32. Examine the battery for the requirements of 3.8.18.”

“d. Energize the battery heaters during vibration. Vibrate the battery along each orthogonal axis (in the axis order shown on figure 3 of the general specification) in accordance with the test levels of figure 3 of the specification sheet. Perform two 15-minute logarithmic sweeps from 5-500-5 hertz. Perform one 5-minute logarithmic sweep from 20-30-20 hertz. Perform a resonance search from 5-500 hertz. Perform resonance dwell testing for 1.5 minutes at each resonance to a maximum of three most severe exhibiting a transmissibility of two or greater. Vibrate the battery for a period of 20 minutes per axis in accordance with level 1 of figure 4 of the specification sheet. Vibrate the battery for a period of 5 minutes per axis level 2 of figure 4 of the specification sheet.”

“e. Constant potential charge the battery in accordance with 4.4.2 at 28.25 ±0.25 volts during vibration. Discharge the battery at 15.0 amperes for 1.0 minute before beginning each axis.”



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4.5.22 Ground storage test. In step 4.5.22.a, insert before the first sentence: "Perform the test of 4.5.32. Examine the battery for the requirements of 3.8.18." In step 4.5.22.d, change "18.0 volts" to "20.0 volts."

4.5.23 Physical integrity at high temperature test. In step 4.5.23.a, insert before the first sentence: "Perform the test of 4.5.32. Examine the battery for the requirements of 3.8.18." In steps 4.5.23.b and 4.5.23.k, change "18.0 volts" to "20.0 volts."

4.5.28 Output performance test. Delete the title and text.

7.7 Add the following paragraphs:

"4.5.28 Output performance test (aircraft emergency loads). This test consists of the following steps.

"a. Perform the test of 4.5.32. Examine the battery for the requirements of 3.8.18. Constant potential charge the battery in accordance with 4.4.2.

"b. Stabilize each sample battery's temperature at the test temperature specified below (see 4.4.1). With each sample still in the temperature chamber at the indicated test temperature, discharge each sample as follows:

"(1) Sample 1 in accordance with step 1 of table II of the specification sheet.

"(2) Sample 2 in accordance with step 2 of table II of the specification sheet.

"(3) Sample 3 in accordance with step 3 of table II of the specification sheet.

"c. Constant potential charge each sample at room temperature for 5 hours in accordance with 4.4.2 (see 4.4.1)."

"d. Examine the battery for the requirements of 3.8.16."

7.8 Modify the following paragraph:

4.5.29 Cold and hot temperature charge test. Delete the text.

7.9 Add the following paragraphs:

4.5.29.1 Half-hour charge test. This test consists of the following steps.



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“a. Perform the test of 4.5.32. Examine the battery for the requirements of 3.8.18. Constant potential charge the battery in accordance with 4.4.2 for 5 hours at room temperature (see 4.4.1).

“b. Stabilize the samples (see 4.4.1) at -18° C (-0°F) for sample 1, 15° C (59° F) for sample 2, and 55° C (131° F) for sample 3. Perform 4.5.29.1.c through 4.5.29.1.g with the battery in the environmental chamber at the specified temperature.

“c. Discharge each sample at the 1.0C-rate to a cutoff voltage of 20.0 volts.

“d. Examine the battery for the requirements of 3.8.17.1.

“e. Stabilize the samples at room temperature (see 4.4.1). Constant potential charge the battery for 5 hours in accordance with 4.4.2.

“f. Stabilize the samples (see 4.4.1) at the temperatures of 4.5.29.1.b. Discharge each sample at the 1.0 C-rate to remove all but 0.75 ampere-hours from each sample as determined by 4.5.29.1.d.

“g. Stabilize the samples (see 4.4.1) at the temperatures of 4.5.29.1.b. Constant potential charge the battery for 30 minutes in accordance with 4.4.2.

“h. Immediately after completing 4.5.29.1.g, place the samples in a room temperature environment (see 4.4.1). Discharge the samples at the 1.0 C-rate to a cutoff voltage of 20.0 volts.

“i. Examine the battery for the requirements of 3.8.17.1.

4.5.29.2 One-hour charge test. This test consists of the following steps.

“a. Perform the test of 4.5.32. Examine the battery of the requirements of 3.8.18. Constant potential charge the battery for 5 hours in accordance with 4.4.2.

“b. Stabilize the battery (see 4.4.1) at -40° C (-40° F), Perform 4.5.29.2.c and 4.5.29.1.d with the battery in the environmental chamber at the specified temperature.

“c. Discharge the battery at the 1.0 C-rate to a cutoff voltage of 20.0 volts.

“d. Stabilize the battery (see 4.4.1) at -40° C (-40° F). Constant potential charge the battery in accordance with 4.4.2 for 1.0 hour.



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"e. Immediately after completing 4.5.29.2.d, place the battery in a room temperature environment (see 4.4.1) and discharge at the 1.0 C-rate to a cutoff voltage of 20.0 volts.

"f. Examine the battery for the requirements of 3.8.17.2.

"4.5.32 Diode resistance test. Measure the resistance of the battery charging diode in the forward direction between battery pin A and battery pin B. Measure the resistance of the battery charging diode in the reverse direction between battery pin A and battery pin B. Determine the ratio of the resistance in the forward direction to that in the reverse direction. Examine the battery for the requirements of 3.8.18.

7.10 Modify the following table:

TABLE III. Capacity performance requirements. In requirements (1), (2), and (3), change "18.0 volts" to "20.0 volts."



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TABLE I. Qualification inspection and conformance inspection.

| Test number | Examinations and tests | Qualification inspection | | | | | Conformance inspection | | |
|---|---|---|---|---|---|---|---|---|---|
| | | Sample number | | | Requirement paragraph | Test paragraph | Group A sample 1 100% | Group B sample number | |
| | | 1 | 2 | 3 | | | | 1 | 2 |
| 1 | INCOMING INSPECTION | X | X | X | 3.8.1 | 4.5.1 | | X | X |
| 2 | VISUAL AND MECHANICAL EXAMINATION | X | X | X | 3.5, 3.6, 3.8.2, table II | 4.5.2 | X | X | X |
| 3 | DIMENSIONS AND WEIGHT | X | X | X | Reqmt 2, 3.8.3 | 4.5.3 | X[1/] | X | |
| 4 | DIELECTRIC STRENGTH AND INSULATION RESISTANCE TEST | X | X | X | 3.7, 3.8.4 | 4.5.4 | X[1/] | X | X |
| 5 | DIODE RESISTANCE TEST | X | X | X | 3.8.18 | 4.5.32 | X | X | X |
| 6 | ELECTRICAL TEST PREPARATION | X | X | X | --- | 4.5.6 | | X[4/] | X[4/] |
| 7 | CAPACITY DISCHARGE | X | X | X | Reqmt 3, 3.8.6, 3.8.18 | 4.5.7 | X | X[4/] | X[4/] |
| 8 | AIRCRAFT EMERGENCY LOADS | X | X | X | 3.8.16, 3.8.18 | 4.5.28 | | | |
| 9 | HALF-HOUR CHARGE TEST | X | X | X | 3.7, 3.8.17.1, 3.8.18 | 4.5.29.1 | | | |
| 10 | ONE-HOUR CHARGE TEST | X | X | X | 3.7, 3.8.17.2, 3.8.18 | 4.5.29.2 | | | |
| 11 | CYCLING TEST | X | | | 3.8.9, 3.8.18 | 4.5.10 | | X[4/] | |
| 12 | CAPACITY DISCHARGE AT 49°C (120°F) | | X | | 3.8.6, 3.8.18, reqmt (2) of table III | 4.5.11 | | | |
| 13 | DISCHARGE WHILE INVERTED | | X | | 3.5.9, 3.8.18, reqmt (5) of table III | 4.5.13 | | | X[4] |
| 14 | ALTITUDE TEST | | X | | 3.5.9, 3.7, 3.8.10, 3.8.18 | 4.5.14 | | | X |
| 15 | SHOCK TEST (BASIC DESIGN) | | | | 3.7, 3.8.10, 3.8.18 | 4.5.15 | | X | |



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TABLE I. Qualification inspection and conformance inspection - Continued.

| Test number | Examinations and tests | Qualification inspection | | | | | Conformance inspection | | |
|---|---|---|---|---|---|---|---|---|---|
| | | Sample number | | | Requirement paragraph | Test paragraph | Group A sample 1 100% | Group B sample number | |
| | | 1 | 2 | 3 | | | | 1 | 2 |
| 16 | TEMPERATURE SHOCK TEST | X | | | 3.7, 3.8.10, 3.8.18 | 4.5.16 | | X | |
| 17 | TEMPERATURE RISE AND FLOAT TEST | | X | | 3.7, 3.8.11, 3.8.18, reqmt (1) of table III | 4.5.17 | | | X[4/] |
| 18 | HIGH RATE DISCHARGE TEST | | x | | 3.8.18, 3.8.19 | 4.5.33 | | X[4/] | X[4/] |
| 19 | BATTERY GAS EMISSION TEST | | X | | 3.8.10, 3.8.12, 3.8.18 | 4.5.18 | | | |
| 20 | VIBRATION TEST | | X | | 3.7, 3.8.10, 3.8.18 | 4.5.19 | | | |
| 21 | HUMIDITY TEST | | X | | 3.7, 3.8.10, 3.8.18 | 4.5.20 | | | X |
| 22 | SALT FOG TEST | X | | | 3.7, 3.8.10, 3.8.18 | 4.5.21 | | X | |
| 23 | GROUND STORAGE TEST | | | X[2/] | 3.8.13.1, 3.8.18 | 4.5.22 | | | |
| 24 | DEEP DISCHARGE RECOVERY TEST | | | X | 3.7, 3.8.13.3 | 4.5.27 | | | |
| 25 | PHYSICAL INTEGRITY AT HIGH TEMPERATURE TEST | X | | | 3.7, 3.8.10, 3.8.18, reqmt (1) of table III | 4.5.23 | | | |
| 26 | SHOCK TEST (CRASH SAFETY) | | | X | 3.8.10, 3.8.15 | 4.5.25 | | | |
| 27 | IMPACT RESISTANCE TEST (NONMETALLIC BATTERY CONTAINER) | | [3/] | | 3.8.14 | 4.5.24 | | | |
| 28 | SHELF LIFE TEST | | | X | 3.8.9, 3.8.13.2 | 4.5.26 | | | |
| 29 | FINAL EXAMINATION | X | X | X | 3.7, 3.8.4, 3.8.18 | 4.5.31 | | X | X |



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NOTES TO TABLE I

1/ See sample information in test paragraph.
2/ Manufacturers requesting qualification to this specification may submit an additional (fourth) test sample to expedite the qualification test program.
3/ For nonmetallic battery container designs, two outer nonmetallic battery containers shall be furnished for inspection.
4/ Requirement paragraph 3.8.18 and test of paragraph 4.5.32 are not required for Group B inspection.

TABLE II. Aircraft emergency load requirements.

| Step | Sample number | Discharge temperature (°C) | Load wattage (watts) | Discharge time (seconds) |
|---|---|---|---|---|
| 1 | 1 | -29 | 360<br>250<br>Rest open circuit<br>360<br>250 | 0.20<br>30<br>60<br><br>0.20<br>30 |
| 2 | 2 | 50 | 360<br>250<br>Rest open circuit<br>360<br>250<br>Rest open circuit<br>360<br>250 | 0.20<br>60<br>60<br><br>0.20<br>60<br>60<br><br>0.20<br>60 |
| 3 | 3 | 0 | 360<br>250<br>Rest open circuit<br>360<br>250<br>Rest open circuit<br>360<br>250 | 0.20<br>45<br>60<br><br>0.20<br>45<br>60<br><br>0.20<br>45 |



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7.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Navy – AS

Preparing activity:
Navy – AS
(Project 6140-2012-003)

Review activity:
DLA – CC

Agent:
Navy - SH

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/quicksearch.



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